

ADACS: Adversarial Domain Adaptation via Cross-domain Mixed Sampling

Alice Banaudi
s347490

Carla Finocchiaro
s337024

Alessandra Marchese
s349536

Leonard Vincent Ramil
s347300

Abstract

Unsupervised domain adaptation in semantic segmentation presents significant challenges, especially when generalizing a segmentation model trained on labeled images from a synthetic (source) domain to unlabeled images in a real (target) domain characterized by distinct visual distributions. Establishing performance baselines by training two representative models—a high-accuracy architecture (DeepLabv2) and a real-time architecture (BiSeNet)—on the Cityscapes dataset, while also exploring the impact of different loss functions. Building upon the Domain Adaptation via Cross-domain Mixing for Semantic Segmentation (DACS) approach, which generates hybrid training samples by combining segments of source images with pseudo-labeled regions from target images, we propose an enhanced methodology that incorporates adversarial attack modules to strengthen the robustness and accuracy of pseudo-labels: ADACS. Experimental validation on the GTAV to Cityscapes benchmark demonstrates significant improvements in mean Intersection-over-Union (mIoU) compared to the original DACS and other state-of-the-art methods, approaching the performance of supervised models trained directly on the target domain, without requiring additional real-world annotations. Integrating our enhanced DACS methodology with other architectures represents a promising avenue to minimize the domain shift gap further.

All code is available at: github.com/LeoRamill/ADACS

1. Introduction

Semantic segmentation [4] plays a crucial role in computer vision as it assigns a specific semantic category to every pixel in an image. Unlike image classification, it offers a more detailed understanding by producing pixel-accurate maps of object categories. This makes it particularly valuable for applications that require precise spatial awareness: like self-driving systems, which have to

be able to distinguish roads, pedestrians, and vehicles at the pixel level. One of the main challenges in semantic segmentation is the reliance on high-quality annotated training data. Achieving precise segmentation results requires large datasets with detailed pixel-level annotations, but creating such datasets is labor-intensive and time-consuming. A practical solution is to use synthetic data, which can be generated in large quantities with automatic labeling. However, domain gaps between synthetic and real-world data often cause models to struggle with generalization. Variations in image characteristics across datasets can cause performance drops when a model trained on one dataset is applied to another, limiting its real-world applicability.

In this work, we evaluate the performance and generalization of semantic segmentation networks in different training scenarios. First, we train a classic segmentation model, DeepLabv2 [1], on the Cityscapes [3] dataset to establish a high-accuracy baseline. Then we train a real-time model, BiSeNet [10], under the same conditions to explore the trade-off between accuracy and efficiency. Next, we investigate the domain shift problem by training BiSeNet on synthetic data from the GTAV [7] dataset and testing it on real images from Cityscapes, experimenting with data augmentation strategies aimed at reducing the domain gap by enhancing the diversity of the training data. Finally, we explore domain adaptation techniques to align source and target feature distributions, enabling better generalization to real-world data. Specifically, we adopt a domain adversarial training approach based on two discriminators [10] that guide the segmentation network to produce domain-invariant predictions, leveraging unlabeled target images. Additionally, we employ the DACS method (Domain Adaptation via Cross-domain Mixing of images and labels) to further improve adaptation by mixing source and target domain data at the pixel level. By combining DACS and adversarial adaptation, our model benefits from stronger supervision on mixed inputs and explicit domain alignment, leading to better performance on unlabeled target images.

2. Related work

2.1. Semantic Segmentation Architectures

A variety of architectures have been proposed for semantic segmentation, often balancing accuracy and computational efficiency. DeepLab [1] is a widely adopted high-accuracy model that utilizes atrous convolutions and Conditional Random Fields (CRFs) to refine predictions. On the other hand, real-time models such as BiSeNet [10] have been developed for scenarios where inference speed is critical. These models achieve competitive accuracy by combining spatial and contextual information through parallel pathways, making them suitable for time-sensitive applications.

2.2. Domain Adaptation

Many adversarial training strategies employ domain discriminators to encourage the segmentation network to learn domain-invariant features [9]. Recent surveys [11] provide a comprehensive overview of deep unsupervised domain adaptation techniques, categorizing methods based on feature alignment, adversarial learning, and discrepancy minimization. These approaches often leverage adversarial training to align feature distributions between source and target domains, improving model robustness to domain changes.

2.3. Mixing Strategies

Mixing strategies have gained prominence in computer vision, particularly in the context of supervised and semi-supervised semantic segmentation. The central idea is to construct new training samples by combining pixel-level information from multiple images. This can enhance generalization by exposing the model to more diverse data distributions.

3. Methodology

In this section, we describe the core components of our approach, which combines a standard semantic segmentation pipeline with advanced domain adaptation techniques. We detail the training and evaluation procedures, the datasets used, and the integration of adversarial learning with the DACS method to improve model generalization across domains.

3.1. DeepLabv2

In the first step, we employ a DeepLab-based semantic segmentation architecture: using a ResNet-101 backbone enhanced with atrous convolutions to maintain spatial resolution and expand the receptive field without shrinking feature maps. To capture multi-scale context, we add an Atrous Spatial Pyramid Pooling (ASPP) classification head, which

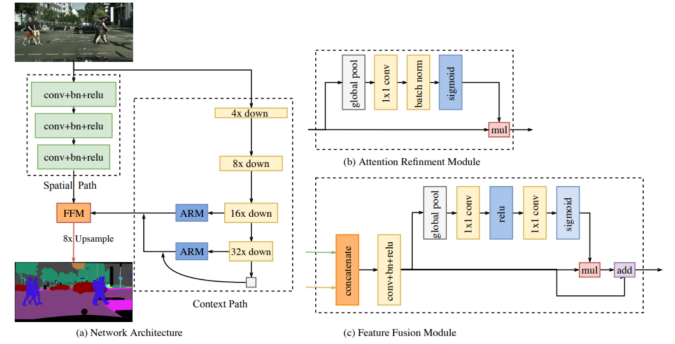


Figure 1. BiSeNet network. Figure from [10]

runs parallel dilated convolutions at different rates and sums their outputs. The final prediction map is bi-linearly up-sampled to match the input resolution. The model is initialized with weights pretrained on ImageNet, except for the ASPP head, which is randomly initialized. During training, we apply a differential learning rate strategy: a base learning rate is used for the backbone, while the classifier head is updated with a higher learning rate.

3.2. BiSeNet

For the following steps of our project we adopted a Bilateral Segmentation Network for Real-time Semantic Segmentation [10]. As shown in Figure 1, the network is composed of two primary branches:

- **Spatial Path:** it consists of three consecutive convolutional blocks with increasing channel depth, each performing down-sampling with stride 2. Thanks to this relatively high resolution, the Spatial Path is able to preserve fine-grained spatial details throughout the network.
- **Context Path:** leverages a pre-trained backbone to extract semantically rich features. Each extracted feature undergoes enhancement via an **Attention Refinement Module (ARM)**. In our experiments, we tested both ResNet-18 and ResNet-101 as backbone networks in order to evaluate the trade-off between segmentation accuracy and computational efficiency.

To combine the outputs from both paths, a **Feature Fusion Module (FFM)** is employed. Concatenates spatial and contextual features, applies a convolutional block, and modulates the combined features using a gated attention mechanism. The model uses deep supervision during training by applying auxiliary supervision to intermediate contextual features prior to fusion. These side outputs are up-sampled and passed through 1x1 convolutions for per-pixel classification.

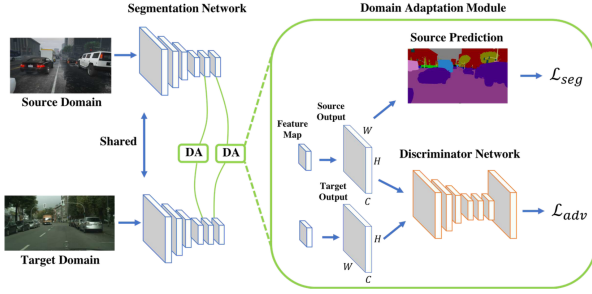


Figure 2. Algorithmic overview for unsupervised adversarial domain adaptation. Figure from [9]

3.3. Unsupervised Adversarial Domain Adaptation

Unsupervised Adversarial Domain Adaptation addresses the challenge of distributional discrepancies between a labeled source domain and an unlabeled target domain, as shown in Figure 2. In semantic segmentation, the goal is to align feature distributions of synthetic images (source domain) with real-world images (target domain), thereby improving generalization to real-world tasks. Formally, given source data (I_s, Y_s) , where I_s is the input image from the source domain and Y_s is the corresponding pixel-wise labels, and unlabeled target data I_t , UDA employs two neural networks: **Generator (Segmentation Network):** $G(I) \rightarrow P$, produces pixel-wise predictions and **Discriminator:** $D(P) \rightarrow [0, 1]$, classifies whether predictions come from the source or target domain. [9] The training objective for the segmentation network G is to minimize the combined loss:

$$\mathcal{L}(I_s, I_t, Y_s) = \mathcal{L}_{seg}(I_s, Y_s) + \lambda_{adv} \mathcal{L}_{adv}(I_t) \quad (1)$$

where:

- The **Segmentation Loss** $\mathcal{L}_{seg}$ is the pixel-wise cross-entropy loss on source images defined as:

$$\mathcal{L}_{seg}(I_s) = - \sum_{h,w} \sum_{c \in C} Y_s^{(h,w,c)} \log \left(P_s^{(h,w,c)} \right) \quad (2)$$

- The **Adversarial Loss** $\mathcal{L}_{adv}$ encourages the target predictions to mimic the predictions of the source domain, defined as:

$$\mathcal{L}_{adv}(I_t) = - \sum_{h,w} \log \left(D(P_t)^{(h,w,1)} \right) \quad (3)$$

The discriminator is optimized separately with a binary cross-entropy loss, distinguishing source predictions from target predictions:

$$\mathcal{L}_d(P) = - \sum_{h,w} (1-z) \log \left(D(P)^{(h,w,0)} \right) + z \log \left(D(P)^{(h,w,1)} \right) \quad (4)$$

where $z = 1$ if the input is from the source domain and $z = 0$ otherwise. [9]

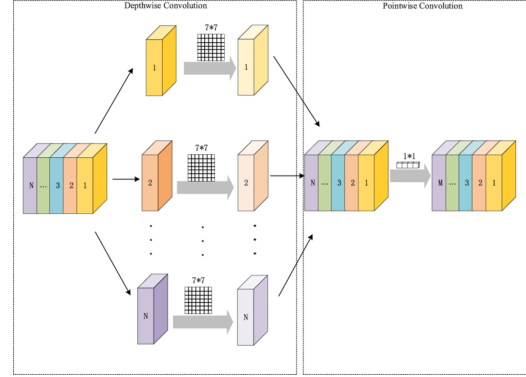


Figure 3. Depthwise Separable Convolutions Steps

3.4. Lightweight Depthwise-Separable Convolutions (DSC)

The adversarial discriminator D follows a standard fully-convolutional design—five successive 4×4 convolutional layers with strides of 2 and channel depths of $\{64, 128, 256, 512, 1\}$ each (except the final) followed by a leaky ReLU ($\alpha = 0.2$) and capped with an upsampling layer to restore spatial resolution. It ordinarily employs 2D convolutions over all input channels; to improve efficiency and lighten the model, these layers can instead be implemented as Depthwise Separable Convolutions (DSC), processing each channel independently (Figure 3). Depthwise-separable convolutions significantly reduce computational complexity and the number of parameters, making convolutional neural networks (CNNs) efficient. [2] A standard convolutional layer with a kernel size $K \times K$, operating over C_{in} input channels and producing C_{out} output channels, requires computational complexity of $O(K^2 \times C_{in} \times C_{out})$. In contrast, a depthwise-separable convolution (DSC) is split into two distinct operations:

1. **Depthwise Convolution:** Applies a single filter per input channel to have complexity: $O(K^2 \times C_{in})$. Formally, given an input $X \in \mathbb{R}^{H \times W \times C_{in}}$, and depthwise convolutional filters $F_d \in \mathbb{R}^{K \times K \times C_{in}}$:

$$X_d^{(h,w,c)} = \left(X * F_d^{(c)} \right) (h, w) \quad (5)$$

where c indexes channels

2. **Pointwise Convolution** (1×1 convolution): Combines outputs from depthwise convolution across channels to have complexity: $O(C_{in} \times C_{out})$. Formally, with pointwise filters $F_p \in \mathbb{R}^{1 \times 1 \times C_{in} \times C_{out}}$:

$$Y^{(h,w,c')} = \sum_{c=1}^{C_{in}} X_d^{(h,w,c)} \cdot F_p^{(1,1,c,c')} \quad (6)$$

The total computational complexity thus reduces significantly to:

$$O(K^2 \times C_{in} + C_{in} \times C_{out}) \quad (7)$$

3.5. DACS: Domain Adaptation via Cross-domain Mixed Sampling

Domain Adaptation via Cross-domain Mixed Sampling (DACS [8]) addresses the domain shift problem in semantic segmentation by introducing a simple yet effective data augmentation strategy that enables semi-supervised learning. DACS constructs hybrid training samples that blend information from both source and target domains, allowing the segmentation model to generalize more effectively to unseen data. As shown in Figure 4, the method operates by sampling a batch of labeled source images and annotations (X_S, Y_S) and a batch of unlabeled target images X_T . The segmentation model f_θ first generates pseudo-labels $\hat{Y}_T$ for the unlabeled target images. A binary mixing mask is then created by selecting a random subset of semantic classes from the source annotation Y_S . This mask is used to merge $\hat{Y}_T$ with Y_S , resulting in a hybrid label Y_M . The same mask is applied to blend the corresponding source and target images into a new input X_M , ensuring spatial alignment with the mixed label. These image-label pairs (X_M, Y_M) are used alongside the original source data during training. The network is trained jointly on the original source pairs and the newly synthesized samples, forming a semi-supervised framework that integrates both labeled and pseudo-labeled data.

This mixing process is inspired by ClassMix [6], an augmentation technique that transfers a subset of semantic regions from one image onto another using class-based binary masks. In the DACS setting, this approach is adapted to mix source and target images, combining the structural clarity of the source with the appearance of the target domain. Although the resulting images are not photorealistic, they retain semantic coherence and serve to bridge the domain gap, as illustrated in Figure 5. Once the mixed and source samples are prepared, training proceeds as in supervised learning: predictions are compared to their respective labels via pixel-wise cross-entropy loss, and parameters are

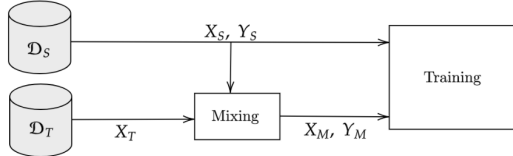


Figure 4. Overview of the DACS framework: a source image X_S and a target image X_T are mixed using a binary mask derived from the source label Y_S . The segmentation network is trained on both the mixed and original source images. Figure from [8].



Figure 5. Example of DACS augmentation: a source image (GTA5) is mixed with a target image (Cityscapes) to create an augmented sample combining regions from both domains. Figure from [8].

updated through backpropagation over N iterations. The overall training objective is defined as:

$$\mathcal{L}_{mix}(\theta) = \mathbb{E} [\mathcal{H}(f_\theta(X_S), Y_S) + \lambda \mathcal{H}(f_\theta(X_M), Y_M)] \quad (8)$$

where $\mathcal{H}$ denotes the cross-entropy loss and λ balances the supervised and unsupervised components.

By generating a continuous stream of semantically consistent mixed samples, DACS constructs an intermediate representation space that narrows the domain gap in a computationally efficient and architecture-agnostic manner.

3.6. Proposed Model: ADACS (Adversarial Domain Adaptation via Cross-domain Mixed Sampling)

The task of unsupervised domain adaptation in semantic segmentation addresses the challenge of generalizing models trained on synthetic source data to real-world target data without relying on additional real-world annotations. While Domain Adaptation via Cross-domain Mixed Sampling (DACS) improves generalization using pseudo-labeling and pixel-based mixing techniques, it can be further enhanced through explicit domain alignment. Therefore, our method, ADACS, integrates DACS with an adversarial component (as shown in Figure 6) to generate more robust and accurate predictions on unlabeled target data.

Mathematically, given a labeled source domain (X_s, Y_s) and an unlabeled target domain X_t , ADACS combines two key methodologies:

- The first is Cross-domain Mixed Sampling, where a binary mask M is generated by randomly selecting half of the semantic classes present in the source labels Y_s
- The second component, Adversarial Domain Adaptation, introduces a discriminator D to distinguish whether predictions originate from the source or target domain.

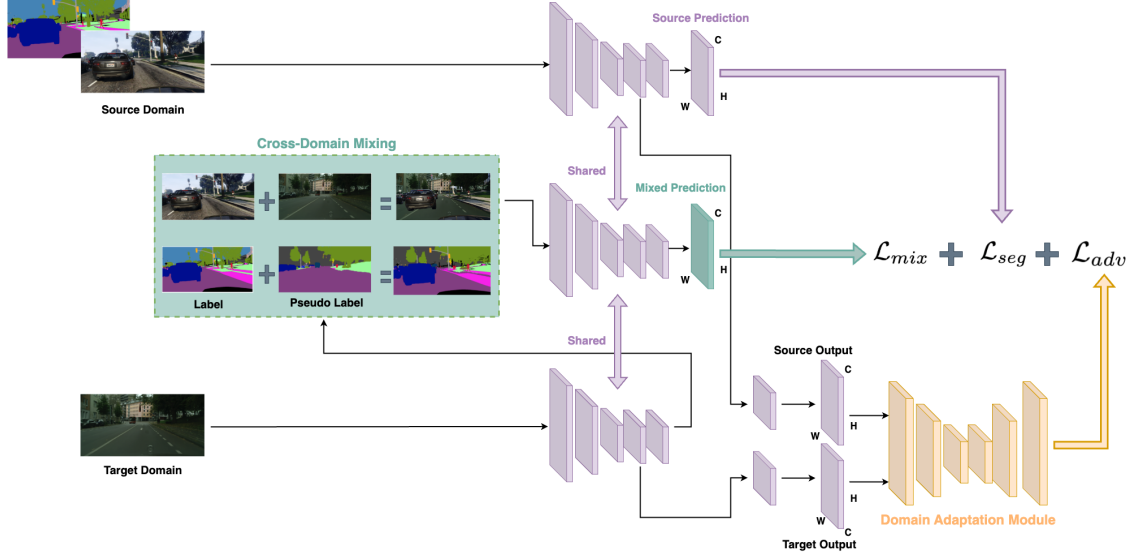


Figure 6. Overview of the ADACS pipeline. A shared segmentation backbone processes both source and target images to produce source predictions and target pseudo-labels. Cross-domain mixing (green) combines source images and ground-truth labels with target images and pseudo-labels via a class-aware mask to generate mixed samples. Simultaneously, source and target predictions are fed to an adversarial domain adaptation module (orange), which distinguishes between domains. The final training objective integrates $\mathcal{L}_{seg}$, $\mathcal{L}_{mix}$, and $\mathcal{L}_{adv}$ for end-to-end domain adaptation

The overall combined objective for ADACS can thus be expressed as:

$$\mathcal{L}_{ADACS}(X_s, X_t, Y_s) = \mathcal{L}_{seg}(X_s, Y_s) + \lambda_{adv} \mathcal{L}_{adv}(X_t) + \lambda_{dacs} \mathcal{L}_{mix} \quad (9)$$

Integrating DACS with adversarial training in ADACS enhances the robustness of pseudo-labels, explicitly aligns feature distributions between domains, and substantially improves the model’s generalization capability. Ultimately, ADACS aims to achieve performance comparable to supervised models trained directly on the target domain, thereby eliminating the requirement for additional manual annotations.

4. Experiments and results

4.1. Datasets

In our work, we rely on two widely adopted datasets for urban scene understanding: Cityscapes and GTAV. These datasets are complementary in nature and are used together to address the challenges posed by domain shift between synthetic and real-world data.

- **Cityscapes** [3]: a real-world dataset collected from various German cities. It comprises 5,000 high-resolution images with fine pixel-level annotations, covering 19 semantic classes relevant to urban environments.

- **GTAV** [7]: a synthetic dataset derived from the video game Grand Theft Auto V. It includes over 24,000 densely annotated frames with the same label set as Cityscapes. Despite being synthetically generated, the images are highly photo-realistic and visually diverse.

4.2. Optimizer and Scheduler

We employ **Stochastic Gradient Descent (SGD)** as our optimizer, which iteratively updates the model parameters θ in the direction opposite to the gradient of the loss $\mathcal{L}$:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t) \quad (10)$$

where η is the learning rate. SGD is preferred for its simplicity and efficiency on large-scale datasets.

For the learning rate schedule, we use a **polynomial decay scheduler**, common in semantic segmentation. At epoch t , the learning rate lr_t is given by:

$$lr_t = lr_0 \left(1 - \frac{t}{T}\right)^{\text{power}} \quad (11)$$

where lr_0 is the initial learning rate, T is the total number of epochs, and power controls the decay. This strategy allows rapid learning at the start and a gradual decrease to aid convergence.

4.3. Losses

The choice of loss function plays a crucial role in the performance of semantic segmentation models, as it directly

influences how well the model learns to distinguish between different classes at the pixel level. Referring to [5], we experimented with several loss functions:

- **Cross-Entropy Loss:** Commonly used for classification, it measures the dissimilarity between the predicted probability distribution p and the true distribution q :

$$\mathcal{L}_{CE} = - \sum_c q_c \log(p_c). \quad (12)$$

- **Dice Loss:** Often used in segmentation to handle class imbalance by measuring overlap between prediction p and ground truth g :

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum pg}{\sum p + \sum g}. \quad (13)$$

- **Combined Cross-Entropy & Dice Loss:** A weighted sum of cross-entropy and Dice losses to leverage benefits of both:

$$\mathcal{L}_{seg} = \mathcal{L}_{CE} + \mathcal{L}_{Dice} \quad (14)$$

4.4. Baseline assessment

To effectively evaluate our domain adaptation methods, we first established fully supervised baselines. We trained two models on the Cityscapes dataset: DeepLabV2, chosen for accuracy, and BiSeNet, selected for real-time efficiency and subsequent adaptation tasks. Both models were optimized via SGD (momentum=0.9, batch size=3), applying a weight decay of 5×10^{-4} for DeepLabV2 and 1×10^{-4} for BiSeNet [1, 10], using a polynomial learning rate scheduler. For BiSeNet, we confirmed that the original initial learning rate of 2.5×10^{-2} provided superior performance compared to a more conservative rate (Figure 7). Results in terms of mean Intersection-over-Union (mIoU) and efficiency metrics (Table 2) indicate that DeepLabV2 with a ResNet-101 backbone achieves 54.35% mIoU, while BiSeNet with the same backbone achieves a comparable accuracy (54.30%) but with significantly greater efficiency. The ResNet-18 variant of BiSeNet demonstrates real-time capability at the cost of moderately lower accuracy, confirming its excellent trade-off between accuracy and computational cost and establishing it as a strong baseline for domain adaptation experiments.

Table 1. Comparison of task-specific architectures on Cityscapes

Model	Backbone	mIoU (%)	Latency	FPS	FLOPs	Params
Task-specific Architectures						
DeepLabV2	ResNet-101	54.35	69.81 (± 0.76)	14.33 (± 0.15)	374.66G	43.90M
BiSeNet	ResNet-18	48.51	14.69 (± 0.47)	68.14 (± 1.81)	25.78G	12.58M
BiSeNet	ResNet-101	54.30	25.64 (± 2.04)	39.08 (± 1.03)	91.92G	50.8M

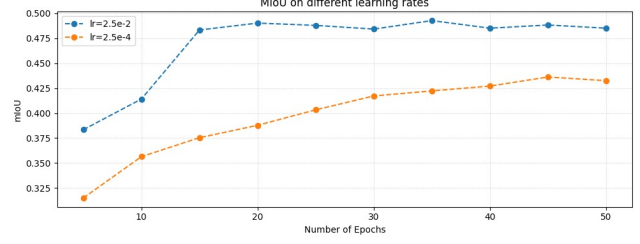


Figure 7. Comparison of learning rate over training epochs for a BiSeNet architecture with a ResNet-18.

4.5. Domain shift

Having established the supervised upper-bound, we now quantify the domain shift by training our BiSeNet model exclusively on the synthetic GTAV dataset and evaluating it on real-world Cityscapes images. Maintaining a consistent training protocol with our BiSeNet-ResNet18 model using a combined Cross-Entropy & Dice loss, this "source-only" setup yields an mIoU of 14.19%, a catastrophic performance degradation from the supervised baseline of 34.32%. This severe performance decline is a well-documented consequence of the "domain gap", a known challenge in synthetic-to-real adaptation tasks such as from GTAV to Cityscapes [11]. To mitigate this gap and improve the model's robustness, we explored several data augmentation strategies applied to the source data. These techniques help the model generalize by simulating variations present in the real world. We evaluated three augmentation sets:

- **aug1:** combination of brightness adjustment (to simulate different condition of light, e.g. day or night) and slight hue shift to simulate variations in color tone to add chromatic variety.
- **aug2:** application of Gaussian blur to mimic image blurriness or out-of-focus effects.
- **aug3:** combination of augmentation 1 and 2

The results of these experiments are summarized in Table 3. The data clearly shows that aug1 provides the most significant performance boost. Concurrently, the use of the combined Cross-Entropy & Dice loss consistently outperforms the standard Cross-Entropy across all augmentation strategies, achieving the best source-only result of 23.80% mIoU with aug1. Based on these findings, we selected aug1 and the combined loss function as the foundational setup for our subsequent domain adaptation experiments.

4.6. Domain Adaptation

For the unsupervised domain adaptation experiments, we adopted the single-level adversarial setting. Two discriminator architectures were evaluated: the Fully Convolutional

Table 2. Model Evaluation results on Cityscapes

Model	Backbone	road	sidewalk	building	wall	fence	pole	light	sign	vegetation	terrain	sky	person	rider	car	truck	bus	train	motorcycle	bicycle	mIoU (%)
DeepLabV2	ResNet-101	95.85	70.85	83.66	34.31	35.26	20.19	32.43	45.94	84.39	50.83	86.82	53.31	28.53	86.29	49.01	50.78	41.92	31.45	53.21	54.35
BiSeNet	ResNet-18	96.05	75.17	83.71	22.00	24.00	22.52	30.90	47.21	85.64	53.49	89.46	53.67	13.87	84.48	18.87	25.26	30.76	15.22	43.3	48.51
BiSeNet	ResNet-101	96.58	75.31	84.69	41.27	36.48	17.43	31.31	44.68	85.77	56.60	88.65	52.24	22.21	85.93	45.50	40.86	33.49	27.65	51.52	54.30

Table 3. Comparison of task-specific architectures on Cityscapes

Model	CrossEntropy	CrossEntropy + Dice
BiSeNet - Aug. 1	21.62	23.80
BiSeNet - Aug. 2	16.07	18.97
BiSeNet - Aug. 3	18.70	19.40

Discriminator (FCD) and the Depthwise Separable Convolution Discriminator (DSC Discriminator). All hyperparameters were set exactly as in the original study [9] under the single-level configuration: the segmentation network was trained with SGD with Nesterov momentum = 0.9, weight decay = 10^{-4} , initial learning rate = 2.5×10^{-4} , decayed by a polynomial schedule with power = 0.9; both FCD and DSC discriminators were trained with the Adam optimizer (learning rate = 10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.99$) using the same decay schedule. The adversarial loss weight λ_{adv} was set to 0.001 in the single-level setting. An analysis of Tab. 4 further highlights the superior performance of the DSC Discriminator compared to the FCD; the DSC model converges to a final $mIoU \approx 0.27$ with stable performance, while FCD stabilizes around $mIoU \approx 0.26$ with greater variability—indicating that DSC offers computational advantages and more reliable convergence during adversarial training.

4.7. DACS’s experiments

To assess DACS, we compared loss functions, backbones, and learning rates. We evaluated the standard cross-entropy loss against a hybrid loss combining cross-entropy and Dice loss, finding that the hybrid improved both segmentation accuracy and training stability; hence, we adopted it for all experiments. Initially using ResNet-18, we switched to ResNet-101 to match the original DACS implementation. For learning rates, we tested 2.5×10^{-4} versus 1×10^{-4} , observing that the higher rate caused unstable training, while the lower rate yielded consistent convergence. Consequently, our final setup uses the combined cross-entropy & Dice loss, ResNet-101 backbone, a learning rate of 1×10^{-4} , and an unsupervised-loss weight of $\lambda = 0.1$.

4.8. ADACS’s experiments and Comparison

In our experiments, we trained ADACS using the same hyperparameters as in the adversarial domain adaptation

and DACS experiments, ensuring consistency and comparability between models. Specifically, we adopted the same discriminator architecture, optimizer configurations, and learning rate schedules.

5. Discussions

The results presented in Table 4 and qualitative observations illustrated in Figure 8 clearly highlight the advantages of our proposed ADACS model compared to previously analyzed approaches, specifically DACS and adversarial adaptation with Depthwise Separable Convolutions (DSC). The ADACS model, which synergistically integrates cross-domain mixing with adversarial training, demonstrates significant superiority in terms of overall mean Intersection-over-Union (mIoU). More specifically, ADACS achieves better generalization on critical classes, consistently outperforming both the DACS and the adversarial adaptation methods individually. For instance, using the ResNet-101 backbone, ADACS reaches an mIoU of 37.01%, compared to 33.21% for the adversarial adaptation with DSC alone, and 27.57% for the DACS model. Additionally, as illustrated in Figure 8, the ADACS model produces more coherent predictions and is less prone to typical errors such as confusing sidewalks with roads, a problem still evident in the DACS model. However, despite these significant improvements, considerable difficulty remains in predicting classes with fewer pixels, such as “bicycle,” “motorcycle,” and “train,” which are largely unpredicted. This highlights the necessity for further improvements to the model. As a future direction, it is therefore advisable to explore a more thorough hyperparameter tuning process, especially focusing on optimizer and learning rate. Optimizing these parameters further could significantly enhance convergence efficiency and the model’s generalization capabilities. Additionally, another promising research avenue is the integration of contrastive learning techniques. These techniques aim to enhance the representation of minority classes by generating more discriminative features through maximizing the similarity between positive examples and minimizing the similarity with negative examples. This approach could effectively mitigate the adverse effects caused by the scarcity of less represented classes, thereby improving the robustness and predictive capability of the model.

Table 4. Segmentation results on the GTA5 $\rightarrow$ Cityscapes task using BiSeNet with various augmentation and domain adaptation strategies.

GTA5 $\rightarrow$ Cityscapes																				
Method	road	sidewalk	building	wall	fence	pole	light	sign	vegetation	terrain	sky	person	rider	car	truck	bus	train	motorcycle	bicycle	mIoU (%)
<i>Backbone Res-Net-18</i>																				
BiSeNet - Domain Shift	22.15	5.13	49.64	1.44	0.26	8.28	3.55	2.60	53.93	2.73	54.69	7.70	0.0	40.24	2.28	0.77	0.0	0.0	0.0	14.19
BiSeNet - Aug. 1	59.44	12.47	65.33	5.52	4.64	9.79	4.81	2.38	74.42	16.43	76.35	13.53	0.01	30.08	6.48	2.64	0.58	0.004	0.0	23.8
BiSeNet - Aug. 2	26.91	4.14	51.94	7.82	2.82	8.45	4.68	2.83	67.61	4.33	67.55	15.76	0.33	21.04	1.40	1.80	0.0	0.0006	0.0	18.97
BiSeNet - Aug. 3	28.85	14.53	53.30	3.378	9.02	7.04	11.35	10.1	69.19	8.89	50.46	23.59	0.57	32.89	0.94	1.89	0.35	0.21	0.0	18.70
BiSeNet - Adv. DA with FCD + Aug. 1	81.53	19.57	68.59	13.10	15.32	6.47	9.68	6.36	64.28	9.16	63.30	25.83	0.76	58.22	7.12	11.15	7.53	4.28	0.0	26.24
BiSeNet - Adv. DA with DSC + Aug.1	79.64	19.65	73.42	13.03	15.02	8.69	16.40	11.63	74.77	20.94	72.79	26.72	3.09	58.90	8.60	15.29	1.34	5.73	0.0	27.36
BiSeNet - DACS + Aug.1	23.43	7.65	70.76	12.43	13.96	7.23	14.66	6.13	75.69	20.90	73.92	25.64	3.39	15.24	6.11	12.11	4.36	1.37	0.0	23.17
BiSeNet - ADACS with DSC + Aug.1	79.85	9.71	67.48	18.17	15.51	7.03	4.88	9.75	76.47	26.42	72.22	29.77	1.89	65.89	16.52	17.55	0.0	0.0	0.0	29.76
<i>Backbone Res-Net-101</i>																				
BiSeNet - Aug. 1	63.58	19.20	65.25	6.62	7.08	11.64	19.36	13.31	79.29	22.75	59.73	33.52	0.0	45.42	19.88	5.75	0.45	0.02	0.0	26.38
BiSeNet - Adv. DA with DSC + Aug.1	82.13	21.43	76.29	14.89	14.99	11.11	24.89	10.23	77.67	21.26	74.31	39.02	2.93	65.77	17.90	17.70	12.71	12.44	0.0	33.21
BiSeNet - DACS + Aug.1	26.75	9.83	60.59	17.98	16.24	10.44	21.65	5.08	78.41	27.11	79.57	35.15	1.82	73.08	20.47	17.46	12.97	9.14	0.0	27.57
BiSeNet - ADACS with DSC + Aug.1	87.51	10.94	78.60	22.78	18.85	11.29	21.37	6.27	80.93	31.32	79.01	40.84	3.92	79.44	25.50	24.73	20.35	12.86	0.0	37.01

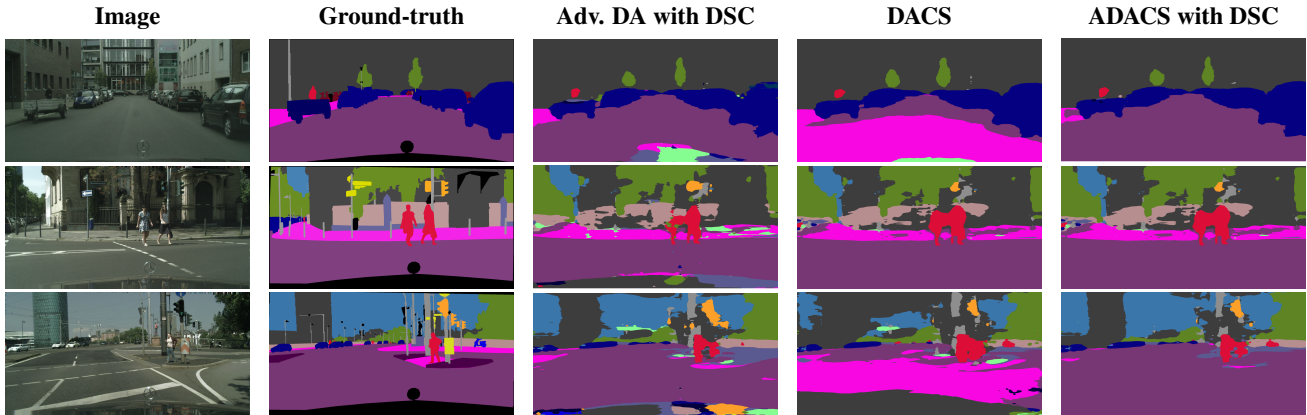


Figure 8. Visual comparison of semantic segmentation outputs for three domain adaptation methods: Adv. DA with DSC, DACS, and the proposed ADACS with DSC. For each example, we show the input image, ground truth annotation, and the corresponding predictions.

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